

인공지능개론

미니프로젝트 #1. Fashion MNIST 데이터셋을 이용한 CNN 이미지 분류

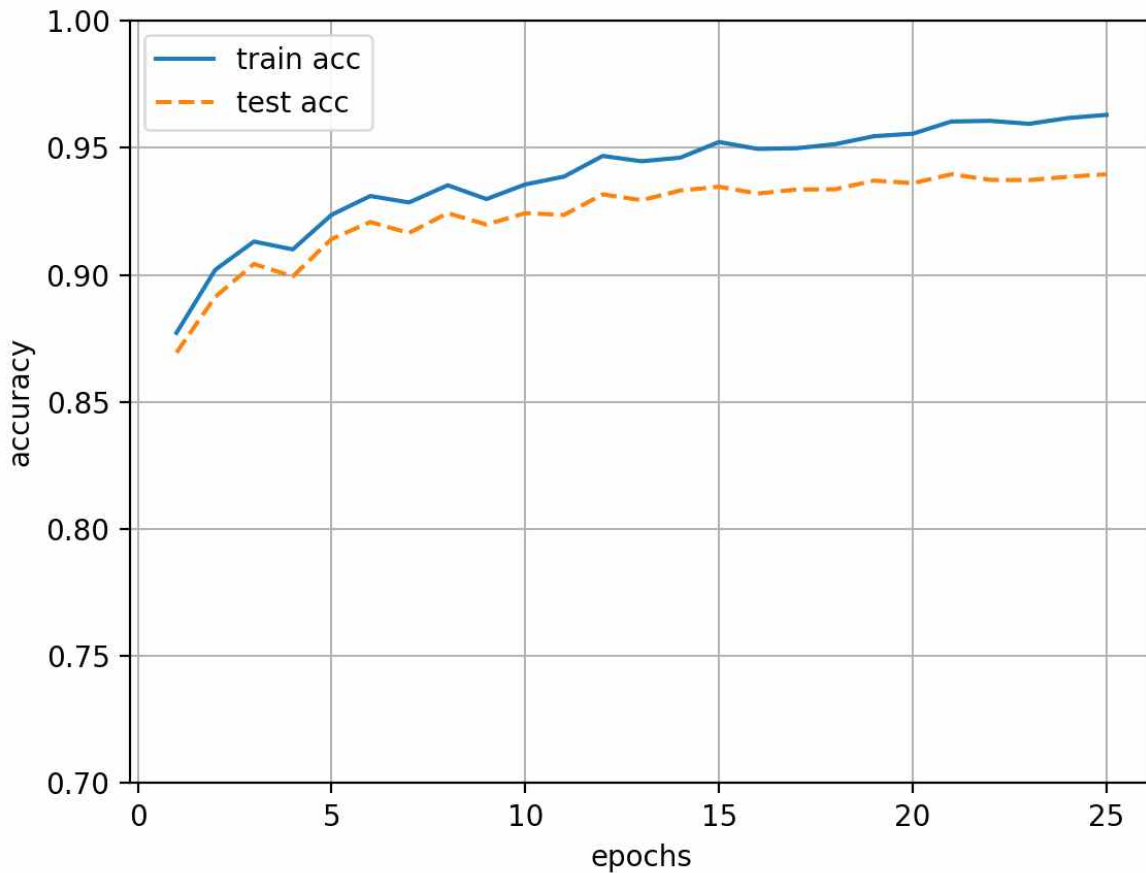
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학번: 202204146, 202204189

1. Pytorch CNN 구현 결과

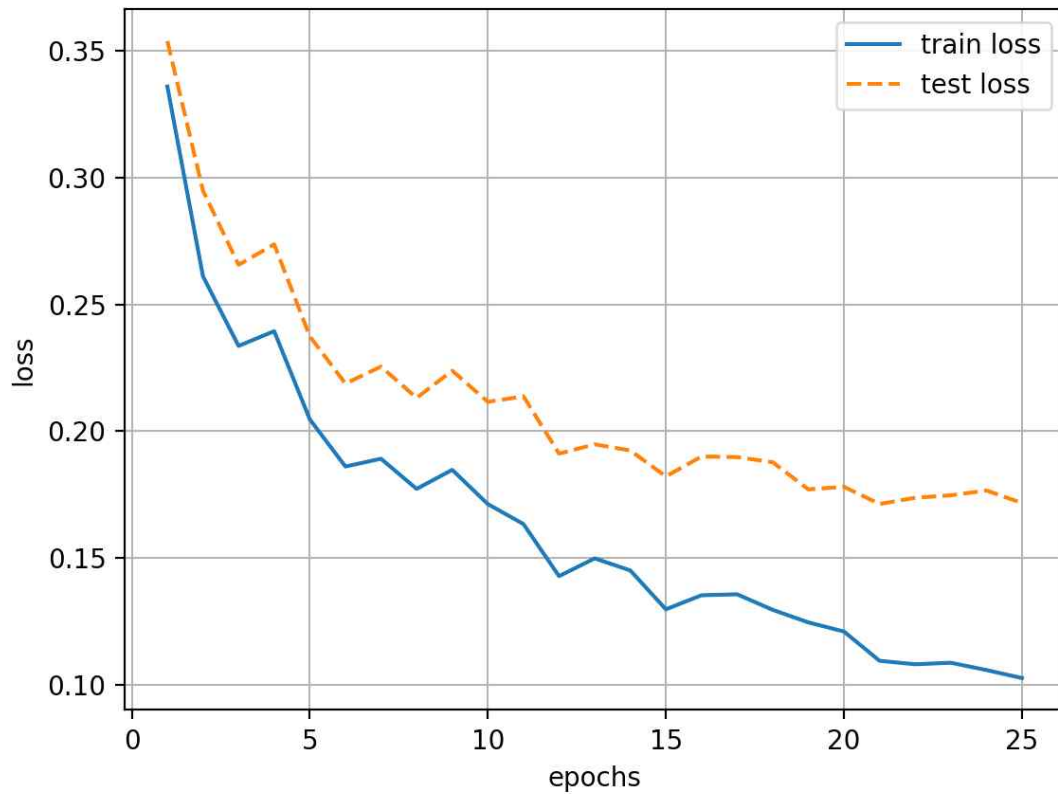
사용장치: NVIDIA Geforce RTX 4060 Ti

Epoch: 25



[그림1] PyTorch CNN의 훈련/테스트 정확도 변화

위 그래프는 PyTorch 기반 CNN 모델의 epoch별 훈련 정확도와 테스트 정확도 변화를 나타낸 것이다. epoch가 증가할수록 훈련 정확도와 테스트 정확도가 함께 상승하는 경향을 보였으며, 최종적으로 훈련 정확도 96.30%, 테스트 정확도 93.96%를 달성하였다.



[그림2] PyTorch CNN의 훈련/테스트 손실 변화

위 그래프는 PyTorch 기반 CNN 모델의 epoch별 훈련 손실과 테스트 손실 변화를 나타낸 것이다. 학습이 진행될수록 손실 값이 전반적으로 감소하였으며, 이는 모델이 Fashion MNIST 데이터의 특징을 점차 학습하고 있음을 보여준다.

최종적으로 테스트 손실도 안정적으로 낮은 값을 유지하여, 모델이 훈련 데이터뿐만 아니라 테스트 데이터에 대해서도 적절한 분류 성능을 보였음을 확인할 수 있다.

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pytorch > ≡ last_20_log.txt
1 Epoch [06/25] Train Loss: 0.1860 Train Acc: 93.11% Test Loss: 0.2188 Test Acc: 92.08% Aug Train Acc: 90.45% Best Test Acc: 92.08%
2 Epoch [07/25] Train Loss: 0.1891 Train Acc: 92.85% Test Loss: 0.2255 Test Acc: 91.65% Aug Train Acc: 90.68% Best Test Acc: 92.08%
3 Epoch [08/25] Train Loss: 0.1772 Train Acc: 93.53% Test Loss: 0.2131 Test Acc: 92.43% Aug Train Acc: 91.16% Best Test Acc: 92.43%
4 Epoch [09/25] Train Loss: 0.1847 Train Acc: 92.98% Test Loss: 0.2239 Test Acc: 91.98% Aug Train Acc: 91.34% Best Test Acc: 92.43%
5 Epoch [10/25] Train Loss: 0.1712 Train Acc: 93.56% Test Loss: 0.2116 Test Acc: 92.43% Aug Train Acc: 91.65% Best Test Acc: 92.43%
6 Epoch [11/25] Train Loss: 0.1633 Train Acc: 93.87% Test Loss: 0.2137 Test Acc: 92.36% Aug Train Acc: 92.53% Best Test Acc: 92.43%
7 Epoch [12/25] Train Loss: 0.1428 Train Acc: 94.68% Test Loss: 0.1911 Test Acc: 93.17% Aug Train Acc: 92.81% Best Test Acc: 93.17%
8 Epoch [13/25] Train Loss: 0.1498 Train Acc: 94.47% Test Loss: 0.1948 Test Acc: 92.94% Aug Train Acc: 93.17% Best Test Acc: 93.17%
9 Epoch [14/25] Train Loss: 0.1450 Train Acc: 94.61% Test Loss: 0.1924 Test Acc: 93.32% Aug Train Acc: 93.19% Best Test Acc: 93.32%
10 Epoch [15/25] Train Loss: 0.1297 Train Acc: 95.23% Test Loss: 0.1821 Test Acc: 93.47% Aug Train Acc: 93.39% Best Test Acc: 93.47%
11 Epoch [16/25] Train Loss: 0.1352 Train Acc: 94.96% Test Loss: 0.1900 Test Acc: 93.20% Aug Train Acc: 93.33% Best Test Acc: 93.47%
12 Epoch [17/25] Train Loss: 0.1356 Train Acc: 94.98% Test Loss: 0.1897 Test Acc: 93.36% Aug Train Acc: 93.42% Best Test Acc: 93.47%
13 Epoch [18/25] Train Loss: 0.1295 Train Acc: 95.14% Test Loss: 0.1877 Test Acc: 93.37% Aug Train Acc: 93.63% Best Test Acc: 93.47%
14 Epoch [19/25] Train Loss: 0.1245 Train Acc: 95.46% Test Loss: 0.1770 Test Acc: 93.71% Aug Train Acc: 93.69% Best Test Acc: 93.71%
15 Epoch [20/25] Train Loss: 0.1209 Train Acc: 95.56% Test Loss: 0.1780 Test Acc: 93.61% Aug Train Acc: 93.69% Best Test Acc: 93.71%
16 Epoch [21/25] Train Loss: 0.1093 Train Acc: 96.04% Test Loss: 0.1712 Test Acc: 93.96% Aug Train Acc: 94.41% Best Test Acc: 93.96%
17 Epoch [22/25] Train Loss: 0.1080 Train Acc: 96.06% Test Loss: 0.1737 Test Acc: 93.74% Aug Train Acc: 94.40% Best Test Acc: 93.96%
18 Epoch [23/25] Train Loss: 0.1086 Train Acc: 95.95% Test Loss: 0.1747 Test Acc: 93.73% Aug Train Acc: 94.50% Best Test Acc: 93.96%
19 Epoch [24/25] Train Loss: 0.1057 Train Acc: 96.17% Test Loss: 0.1766 Test Acc: 93.86% Aug Train Acc: 94.79% Best Test Acc: 93.96%
20 Epoch [25/25] Train Loss: 0.1025 Train Acc: 96.30% Test Loss: 0.1716 Test Acc: 93.96% Aug Train Acc: 94.67% Best Test Acc: 93.96%

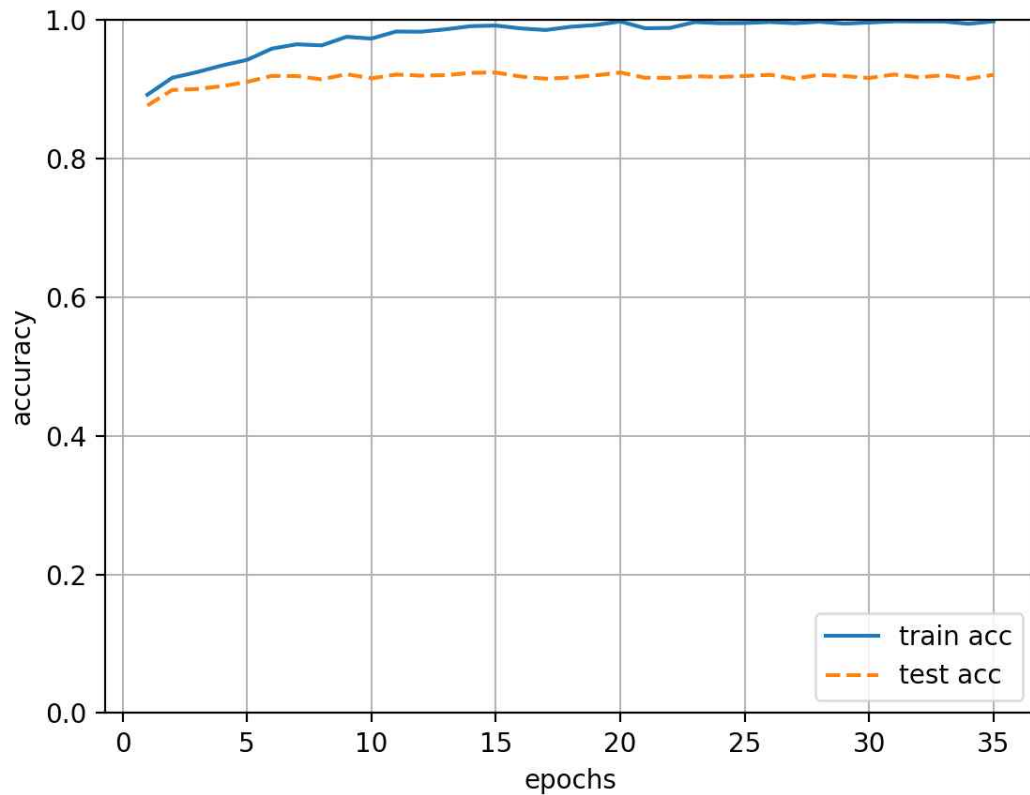
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[그림3] PyTorch CNN 학습 마지막 20줄 로그

마지막 학습 로그를 통해 최종 훈련 정확도와 테스트 정확도가 각각 96.30%, 93.96%로 도달했음을 확인하였다.

최종 훈련 정확도: 96.30%
 최종 테스트 정확도: 93.96%
 최고 테스트 정확도: 93.96%

2. Scratch CNN 학습 결과

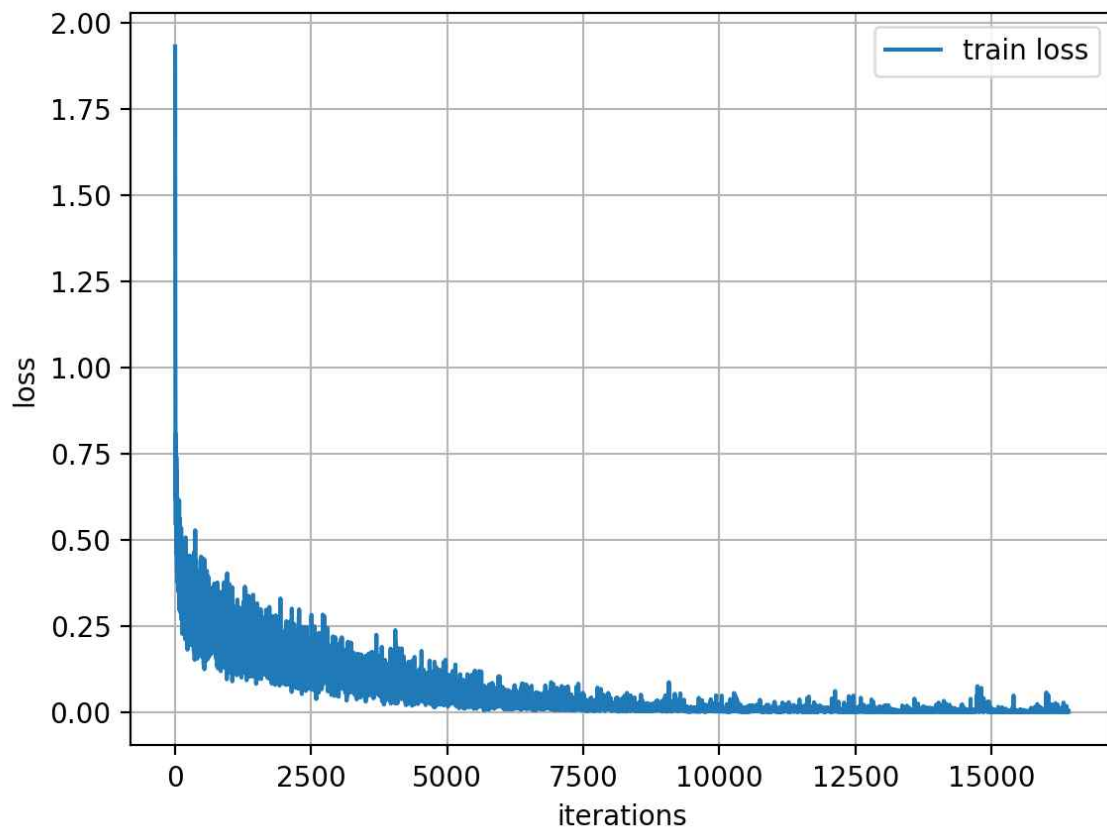


[그림4] Scratch CNN의 훈련 정확도와 테스트 정확도 변화

위 그래프는 교재 CNN 소스코드를 응용한 Scratch 모델의 epoch별 훈련 정확도와 테스트 정확도 변화를 나타낸 것이다.

Scratch 구현에서는 PyTorch, TensorFlow, Keras와 같은 기존 딥러닝 프레임워크 API를 사용하지 않고, 교재 『밑바닥부터 시작하는 딥러닝』의 CNN 관련 소스코드를 응용하였다. 사용한 주요 구성 요소는 im2col, col2im, Convolution, Pooling, Relu, Affine, SoftmaxWithLoss, Adam optimizer이다.

Fashion MNIST는 숫자 MNIST보다 클래스 간 형태가 유사한 경우가 많기 때문에, 기본 SimpleConvNet 구조를 그대로 사용하지 않고 합성곱 계층을 추가하여 특징 추출 능력을 높였다.



[그림5] Scratch CNN의 훈련 손실과 테스트 손실 변화

위 그래프는 Scratch CNN 모델의 학습 과정에서 손실 값이 변화하는 모습을 나타낸 것이다. 학습이 진행될수록 훈련 손실이 감소하였으며, 이를 통해 모델이 Fashion MNIST 데이터셋의 특징을 학습했음을 확인할 수 있다.

```
scratch > ≡ last_20_log.txt
1 Epoch [16/35] Train Loss: 0.0200 Train Acc: 98.79% Test Acc: 91.84% Best Test Acc: 92.42% Epoch Time: 257.6s Total Time: 67.0min
2 Epoch [17/35] Train Loss: 0.0162 Train Acc: 98.56% Test Acc: 91.53% Best Test Acc: 92.42% Epoch Time: 257.4s Total Time: 71.3min
3 Epoch [18/35] Train Loss: 0.0136 Train Acc: 99.02% Test Acc: 91.70% Best Test Acc: 92.42% Epoch Time: 258.0s Total Time: 75.6min
4 Epoch [19/35] Train Loss: 0.0110 Train Acc: 99.28% Test Acc: 92.01% Best Test Acc: 92.42% Epoch Time: 269.3s Total Time: 80.1min
5 Epoch [20/35] Train Loss: 0.0114 Train Acc: 99.79% Test Acc: 92.42% Best Test Acc: 92.42% Epoch Time: 337.3s Total Time: 85.7min
6 Epoch [21/35] Train Loss: 0.0072 Train Acc: 98.81% Test Acc: 91.66% Best Test Acc: 92.42% Epoch Time: 242.8s Total Time: 89.8min
7 Epoch [22/35] Train Loss: 0.0086 Train Acc: 98.86% Test Acc: 91.66% Best Test Acc: 92.42% Epoch Time: 253.9s Total Time: 94.0min
8 Epoch [23/35] Train Loss: 0.0070 Train Acc: 99.70% Test Acc: 91.89% Best Test Acc: 92.42% Epoch Time: 247.4s Total Time: 98.2min
9 Epoch [24/35] Train Loss: 0.0057 Train Acc: 99.54% Test Acc: 91.78% Best Test Acc: 92.42% Epoch Time: 242.3s Total Time: 102.2min
10 Epoch [25/35] Train Loss: 0.0050 Train Acc: 99.56% Test Acc: 91.92% Best Test Acc: 92.42% Epoch Time: 241.3s Total Time: 106.2min
11 Epoch [26/35] Train Loss: 0.0059 Train Acc: 99.71% Test Acc: 92.09% Best Test Acc: 92.42% Epoch Time: 240.6s Total Time: 110.2min
12 Epoch [27/35] Train Loss: 0.0049 Train Acc: 99.54% Test Acc: 91.53% Best Test Acc: 92.42% Epoch Time: 241.1s Total Time: 114.3min
13 Epoch [28/35] Train Loss: 0.0045 Train Acc: 99.75% Test Acc: 92.08% Best Test Acc: 92.42% Epoch Time: 241.6s Total Time: 118.3min
14 Epoch [29/35] Train Loss: 0.0031 Train Acc: 99.48% Test Acc: 91.91% Best Test Acc: 92.42% Epoch Time: 262.8s Total Time: 122.7min
15 Epoch [30/35] Train Loss: 0.0043 Train Acc: 99.64% Test Acc: 91.63% Best Test Acc: 92.42% Epoch Time: 258.7s Total Time: 127.0min
16 Epoch [31/35] Train Loss: 0.0031 Train Acc: 99.81% Test Acc: 92.14% Best Test Acc: 92.42% Epoch Time: 259.3s Total Time: 131.3min
17 Epoch [32/35] Train Loss: 0.0063 Train Acc: 99.80% Test Acc: 91.73% Best Test Acc: 92.42% Epoch Time: 261.3s Total Time: 135.7min
18 Epoch [33/35] Train Loss: 0.0023 Train Acc: 99.79% Test Acc: 92.03% Best Test Acc: 92.42% Epoch Time: 276.1s Total Time: 140.3min
19 Epoch [34/35] Train Loss: 0.0022 Train Acc: 99.45% Test Acc: 91.53% Best Test Acc: 92.42% Epoch Time: 270.6s Total Time: 144.8min
20 Epoch [35/35] Train Loss: 0.0044 Train Acc: 99.81% Test Acc: 92.09% Best Test Acc: 92.42% Epoch Time: 272.5s Total Time: 149.4min
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[그림6] Scratch CNN 학습 마지막 20줄 로그

위 로그는 Scratch CNN 모델의 마지막 학습 결과를 나타낸 것이다. 최종 훈련 정확도 99.81%, 최종 테스트 정확도 92.09%, 최고 테스트 정확도 92.42%를 기록하였다.

Scratch CNN 학습 결과

Epoch: 35

최종 훈련 정확도: 99.81%

최종 테스트 정확도: 92.09%

최고 테스트 정확도: 92.42%